

Project : Artificial Intelligence for Robotics

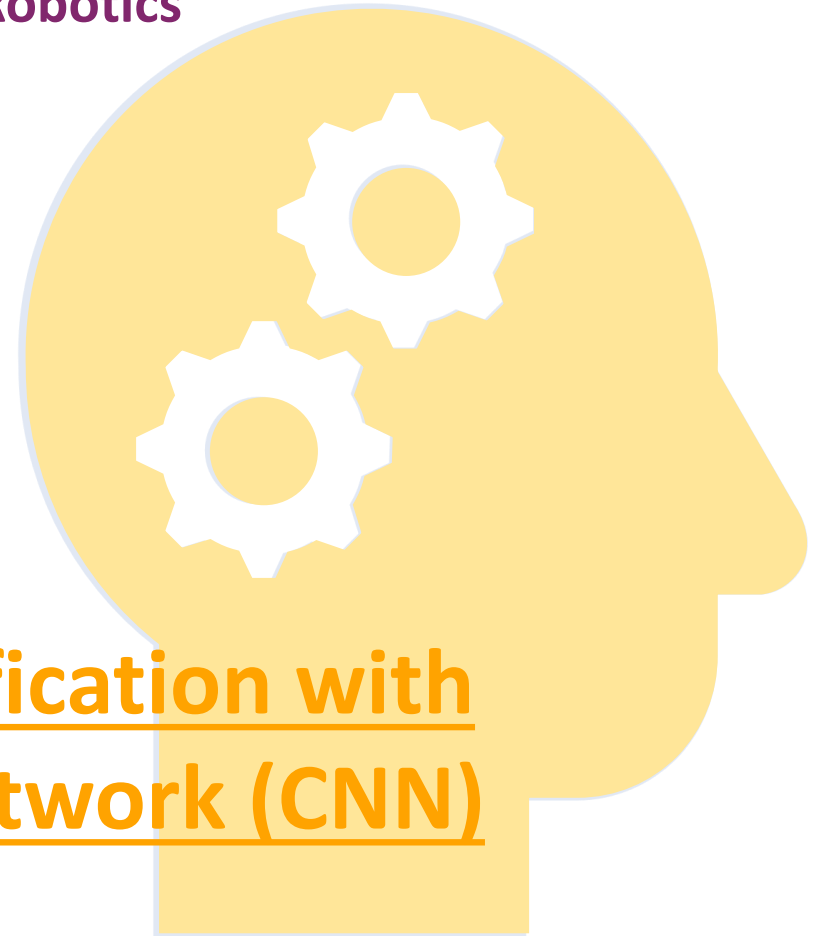


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Supervised Image Classification with Convolutional Neural Network (CNN)





Plan

- Introduction
- Development of a first model using CNN
- Transfer learning: Comparison of KNN algorithm vs final dense layer
- Ensemble method
- Conclusion and outlook

Introduction

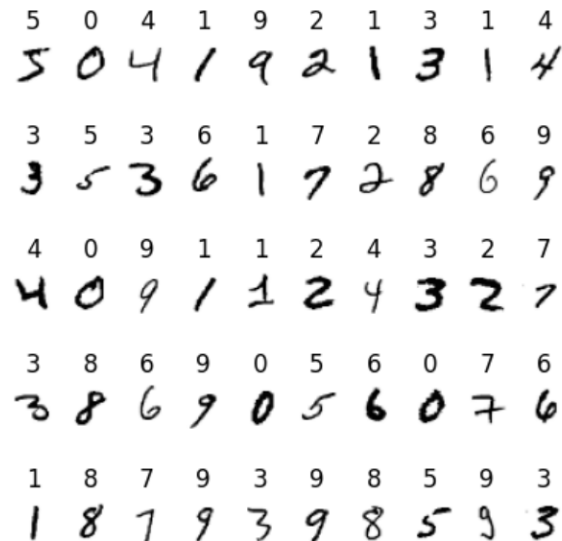
OBJECTIVE:

Create a model capable of recognizing digits or objects using two datasets

Datasets:

- **MNIST:** 70,000 images of handwritten digits (0 to 9)

Des images MNIST



- **Fashion MNIST:** contains images of clothing items in 10 classes.

Des images Fashion MNIST



Development of a first model

using CNN_(1/2)

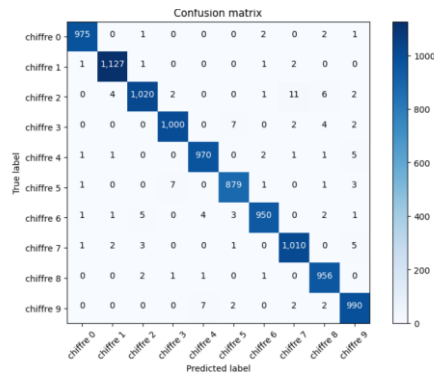
- Convolution layer
- Pooling
- Dense layers

Layer (type)	Output Shape	Param #
conv2d_12 (Conv2D)	(None, 24, 24, 16)	416
max_pooling2d_8 (MaxPooling2D)	(None, 12, 12, 16)	0
conv2d_13 (Conv2D)	(None, 10, 10, 32)	4,640
max_pooling2d_9 (MaxPooling2D)	(None, 5, 5, 32)	0
conv2d_14 (Conv2D)	(None, 3, 3, 32)	9,248
flatten_4 (Flatten)	(None, 288)	0
extraction100 (Dense)	(None, 100)	28,900
extraction40 (Dense)	(None, 40)	4,040
dense_4 (Dense)	(None, 10)	410

Development of a first model using CNN_(2/2)

Results for MNIST:

- Accuracy: 98%



P=9,T=4 P=3,T=5 P=1,T=6 P=4,T=9 P=4,T=1 P=0,T=8 P=2,T=3 P=5,T=9 P=7,T=9 P=7,T=9

4 5 6 9 1 0 3 9 9 9

P=2,T=1 P=9,T=5 P=7,T=9 P=6,T=2 P=2,T=7 P=4,T=6 P=2,T=7 P=3,T=5 P=2,T=8 P=2,T=7

2 5 9 2 7 6 7 0 0 7

P=6,T=4 P=2,T=7 P=4,T=0 P=2,T=1 P=6,T=4 P=2,T=7 P=2,T=9 P=7,T=1 P=3,T=7 P=5,T=3

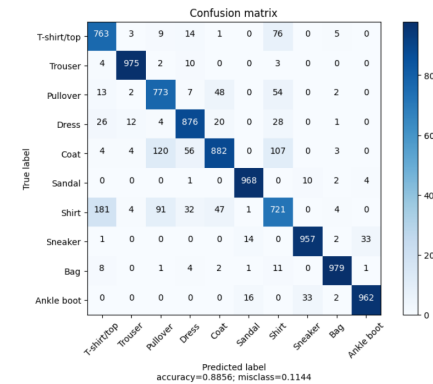
4 7 0 1 4 7 9 1 7 3

P=1,T=7 P=2,T=3 P=2,T=8 P=3,T=9 P=7,T=1 P=3,T=8 P=9,T=8 P=3,T=8 P=5,T=3 P=4,T=9

7 3 2 9 1 8 8 8 3 4

Results for Fashion MNIST:

- Accuracy: 89%



P=4,T=3 P=6,T=4 P=7,T=9 P=4,T=6 P=6,T=0 P=4,T=2 P=6,T=0 P=6,T=0 P=6,T=2 P=6,T=4



P=6,T=0 P=0,T=2 P=4,T=6 P=6,T=0 P=6,T=4 P=6,T=4 P=6,T=0 P=6,T=2 P=7,T=9 P=4,T=6



P=9,T=5 P=4,T=2 P=0,T=3 P=2,T=4 P=6,T=4 P=4,T=2 P=6,T=8 P=2,T=4 P=4,T=6 P=6,T=4



Transfer learning: Comparison of KNN algorithm vs final dense layer^(1/2)

METHOD 1: KNN

Advantages	Disadvantages
- Simplicity of implementation - Effective if the data are well separated in the space	Limited performance on large datasets (MNIST)

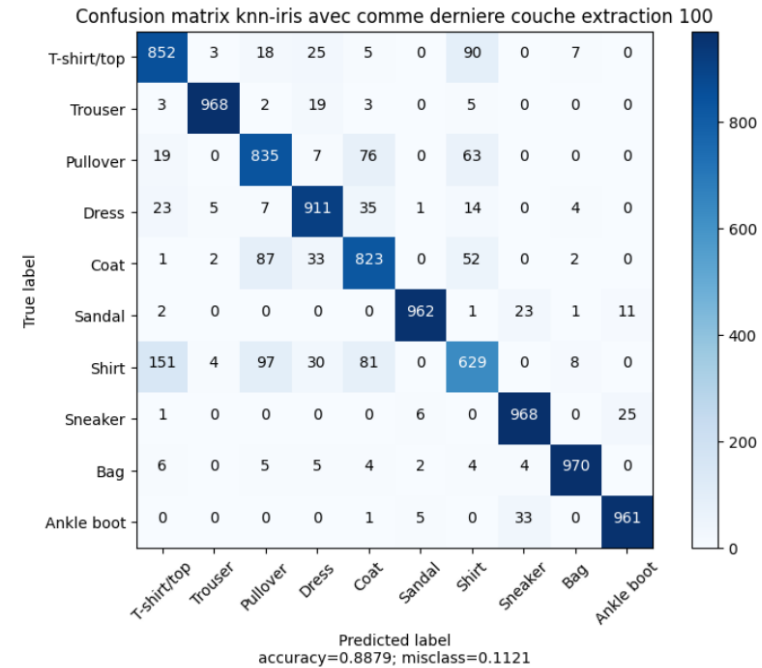
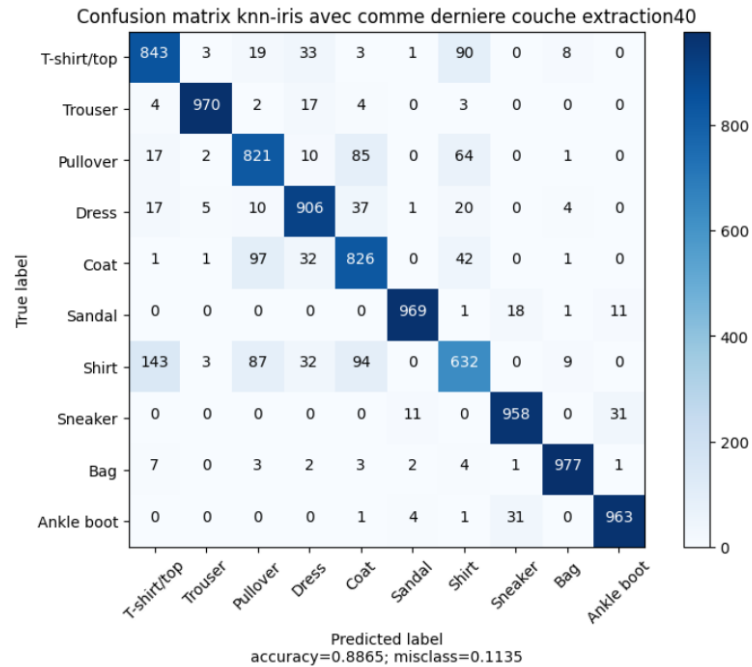
```
# construction du modèle
nb_voisins=10 #choix du paramètre k=nombre de voisins
model = knn.KNeighborsClassifier(n_neighbors=nb_voisins) # Remarque : par défaut n_neighbors=5, ici réglé à nb_voisins.

model.fit(XtrainFeatures,Ytrain)
Ypred=model.predict(XtestFeatures)
```

METHOD 2 : Last dense layer

Part of a model that has already learned to extract and associate complex features with classes or values.

Transfer learning: Comparison of KNN algorithm vs final dense layer^(2/2)



Ensemble method (1/6)

OBJECTIVE: Improving of results

Model 1: Initial Network

Model 2: Sigmoid activation + dropout layer

Model 3: ReLU activation + removal of a layer + dropout

Ensemble method (1/5)

Layer (type)	Output Shape	Param #
conv2d_39 (Conv2D)	(None, 24, 24, 16)	416
max_pooling2d_26 (MaxPooling2D)	(None, 12, 12, 16)	0
conv2d_40 (Conv2D)	(None, 10, 10, 32)	4,640
dropout_19 (Dropout)	(None, 10, 10, 32)	0
max_pooling2d_27 (MaxPooling2D)	(None, 5, 5, 32)	0
conv2d_41 (Conv2D)	(None, 3, 3, 32)	9,248
flatten_13 (Flatten)	(None, 288)	0
extraction100 (Dense)	(None, 100)	28,900
dropout_20 (Dropout)	(None, 100)	0
extraction40 (Dense)	(None, 40)	4,040
dense_12 (Dense)	(None, 10)	410

Layer (type)	Output Shape	Param #
conv2d_45 (Conv2D)	(None, 24, 24, 16)	416
max_pooling2d_30 (MaxPooling2D)	(None, 12, 12, 16)	0
conv2d_46 (Conv2D)	(None, 10, 10, 32)	4,640
dropout_23 (Dropout)	(None, 10, 10, 32)	0
max_pooling2d_31 (MaxPooling2D)	(None, 5, 5, 32)	0
conv2d_47 (Conv2D)	(None, 3, 3, 32)	9,248
flatten_15 (Flatten)	(None, 288)	0
dropout_24 (Dropout)	(None, 288)	0
extraction100 (Dense)	(None, 100)	28,900
dense_13 (Dense)	(None, 10)	1,010

Ensemble Method (2/6)

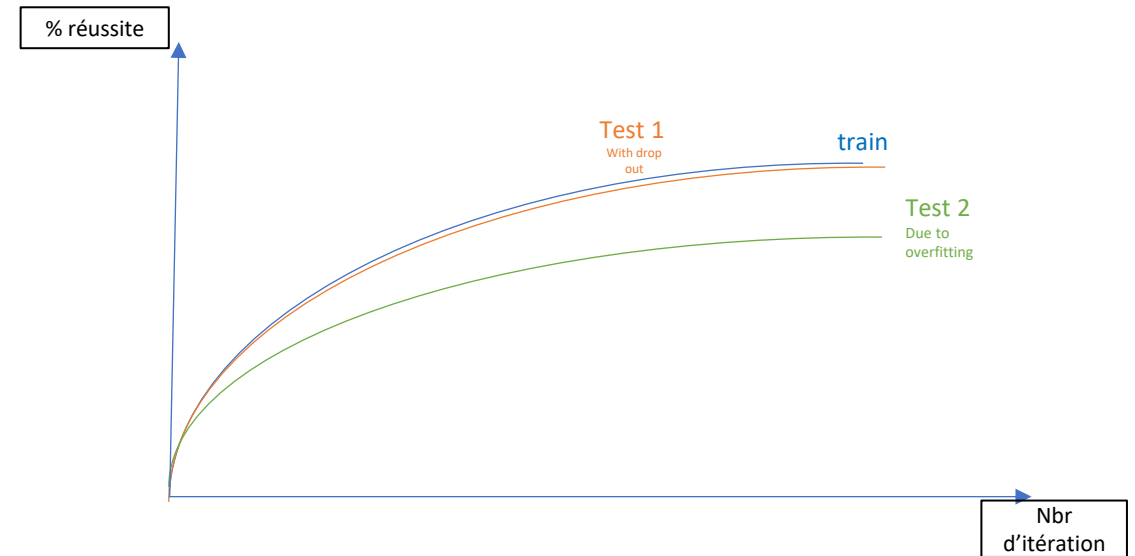
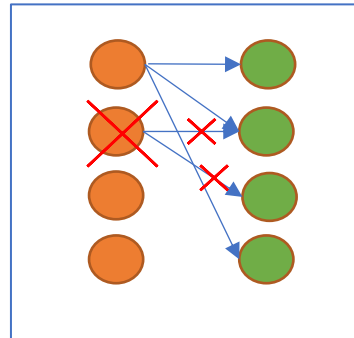
Sigmoid function :

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

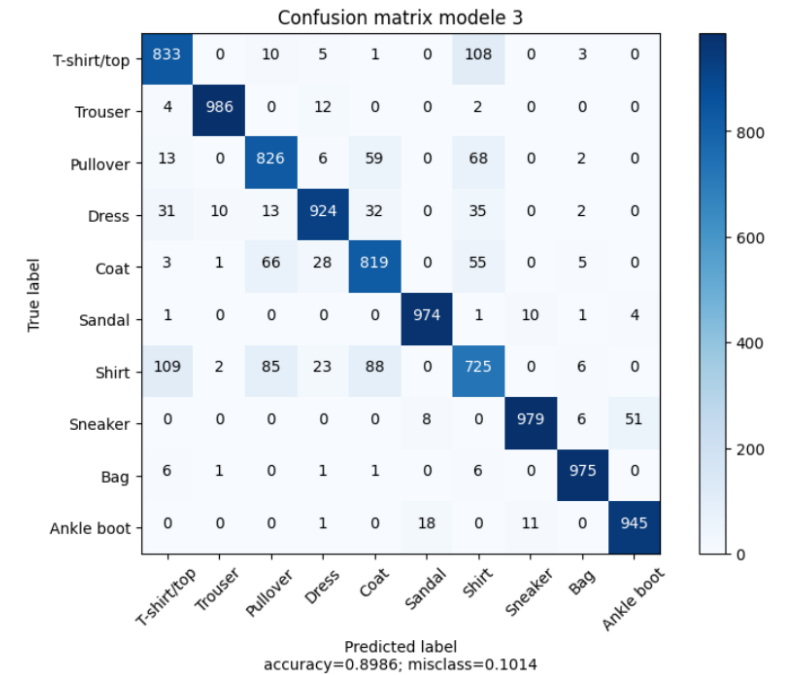
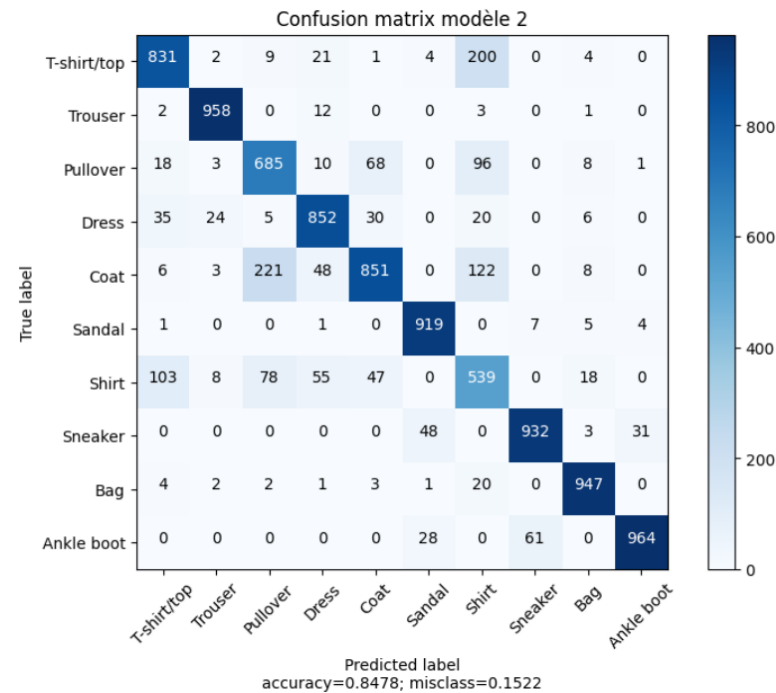
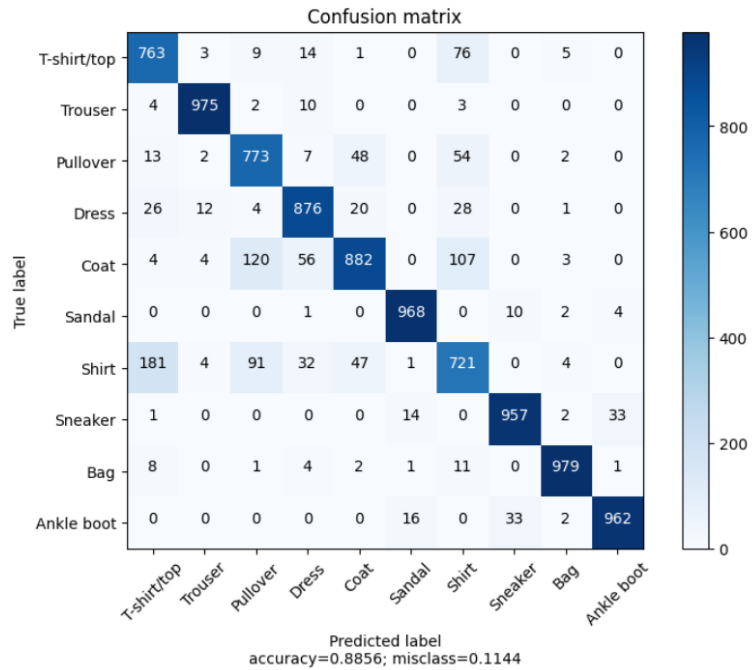
ReLU function

$$f(x) = \max(0, x)$$

DropOut function

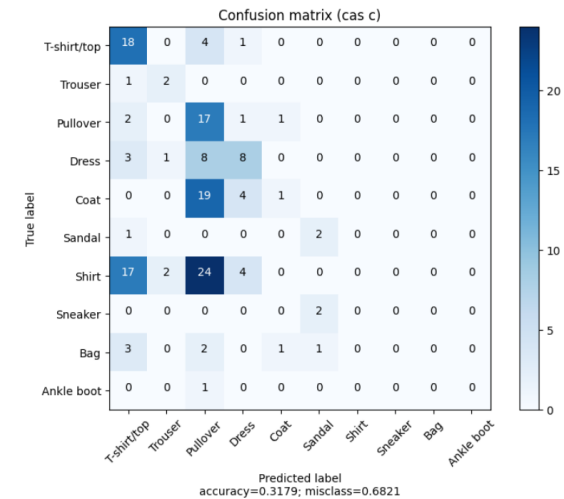
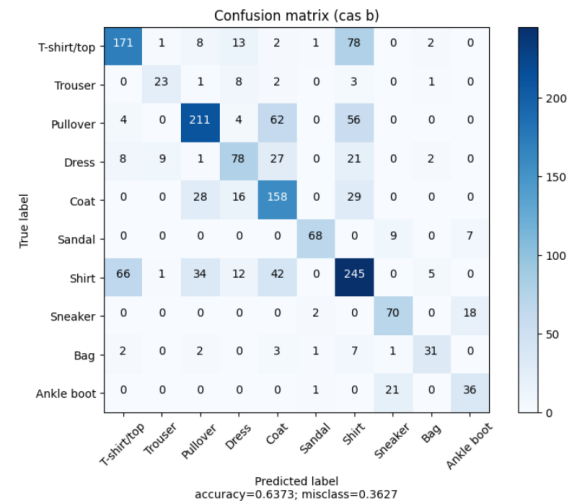
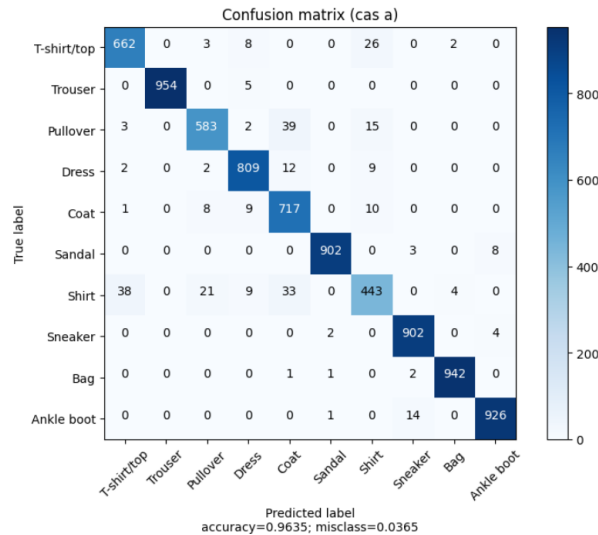


Ensemble method_(3/6)



Aggregation of predictions

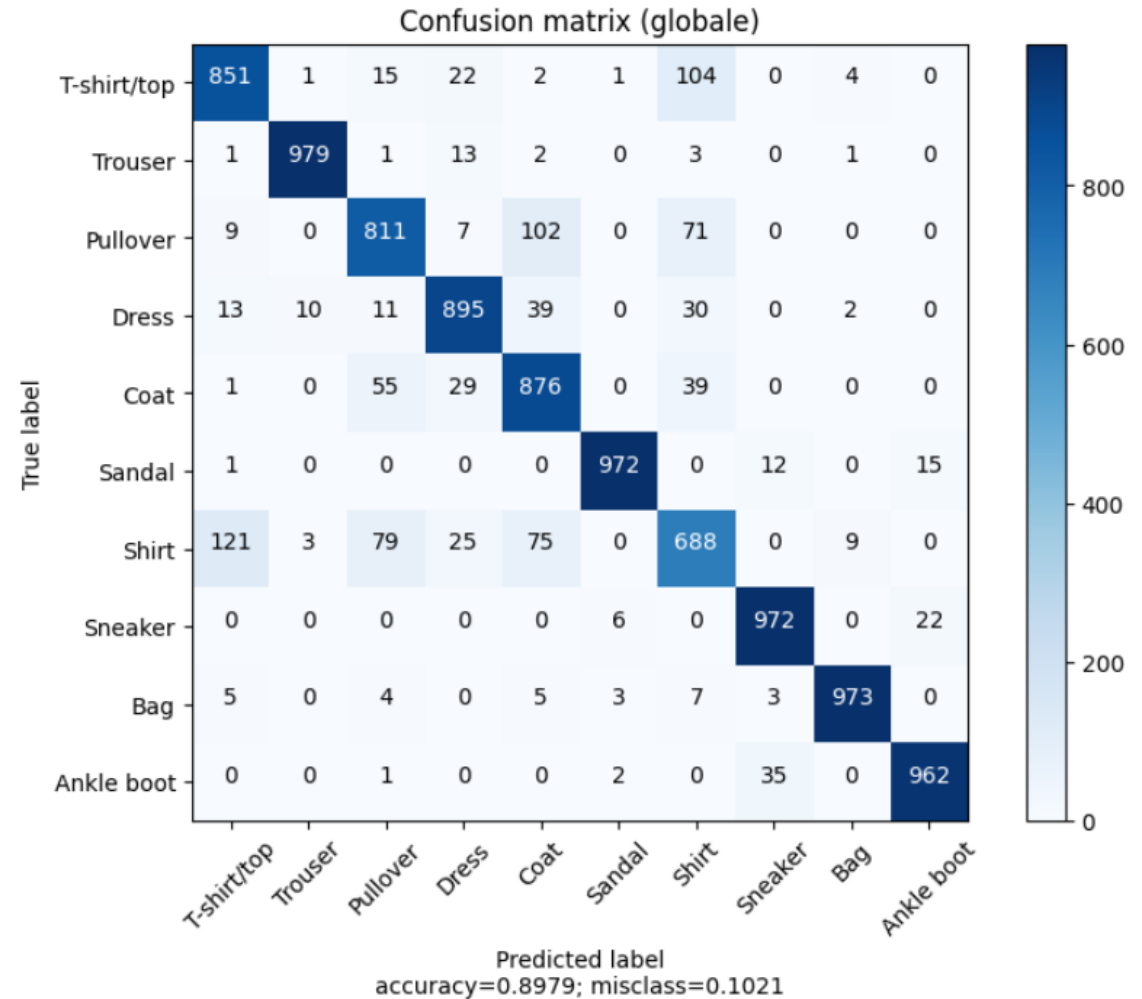
- 1) All models predict the same class.
- 2) Two out of three predict the same class.
- 3) All three predict differently.



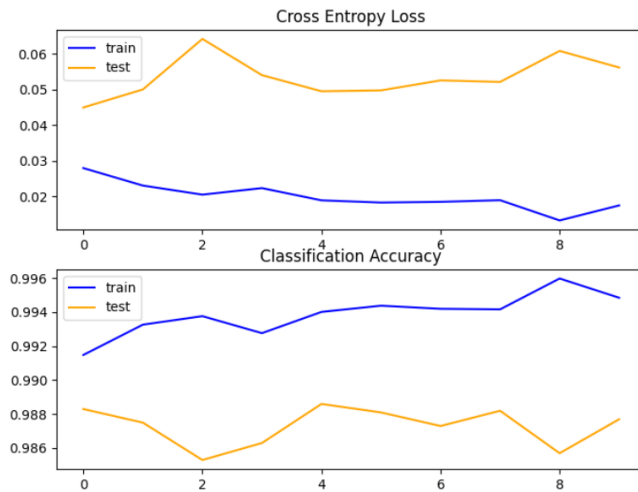
Ensemble Method (5/6)

Improves accuracy.

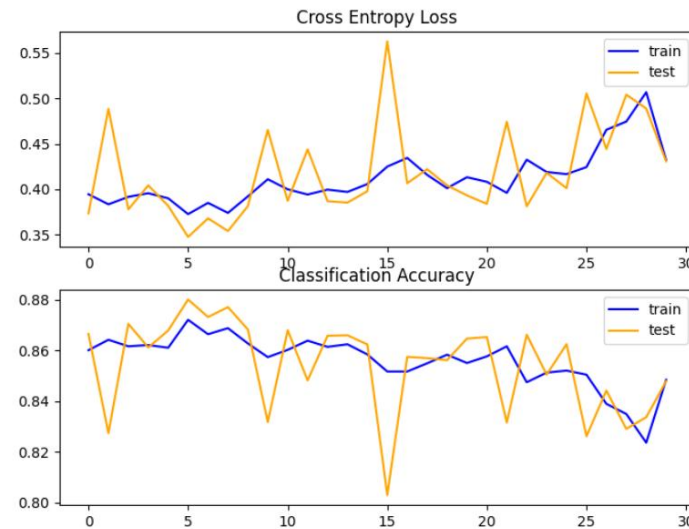
Note: No significant difference in accuracy observed (due to the use of method 2).



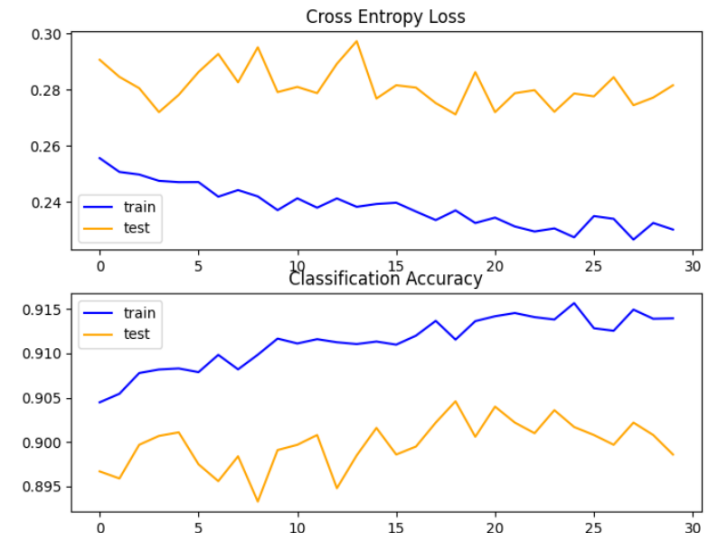
Ensemble Method (6/6)



Pattern 1



Pattern 2



Pattern 3

Conclusion

- Convolutional networks are highly effective for datasets like MNIST.
- Ensemble methods can improve robustness.

WHAT NEXT?

- Increase the amount of data to make the models even more robust.
 - Test with a larger number of networks for aggregation.
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